Main Objective To produce time series of orbital images of a 3U CubeSat labelled with pose ground truth data that represent trajectories relevant for trajectory and pose estimation research.

Concept of Operations In the Demonstration phase:

1. The CUBICS team defines the Demonstration Trajectory (DT), and programs it on the OBC.
   (a) The CUBICS team specifies the pose for each DT step. They could automatically generate them from a proximity operations navigation algorithm, for example.
   (b) The CUBICS team specifies the timestamp for each DT step in relative time from the first pose. They could choose these values based on the latency of a navigation algorithm.
   (c) The CUBICS team specifies the space illumination conditions, space phenomena, and CubeSat orbit position during which the DT needs to be executed. They determine the resulting start time of the trajectory in absolute time.
2. The CubeSat becomes operational, and continuously checks if the current time matches the DT start time.
3. If the start time is matched and the available energy is not sufficient, the CubeSat downlinks its health status and a request for a new start time, then awaits a response to retry execution.
4. If the start time is matched and the available energy is sufficient, the CubeSat deploys the Camera Payload. The Camera Payload navigates to the first pose of the DT and captures an image of the CubeSat. This image is annotated with the pose ground truth and the absolute timestamp. CubeSat health data is also collected for every trajectory pose.
5. The Camera Payload navigates to the next pose in the trajectory within the allowed delay, and repeats the data collection process. Once the DT has been completed, the CubeSat compresses the dataset and downlinks it.
6. The CUBICS team analyzes the DT dataset to check data quality and completeness, and analyze the absolute timestamps to determine whether the DT was actually executed during the intended illumination conditions, space phenomena, and CubeSat orbit position. They publish the dataset.
7. The CUBICS team refines the method to determine the start time of a trajectory from desired illumination conditions, space phenomena, and CubeSat orbit position, and the method to determine how much energy is required to execute a trajectory.

We can call the pose a ground truth because it has to be measured accurately so that the Camera Payload can determine when it has successfully reached the target pose. It the pose isn't known it can't be verified that the predefined trajectory is being respected, and so the value of the data is compromised.

In the Laboratory phase the CubeSat operates in the same sequence, but executes trajectories proposed by external stakeholders that are uplinked after being approved by the CUBICS team and waiting in a queue. If a trajectory cannot be executed due to a lack of energy, the stakeholder is given a time window to propose another start time before they are sent to the back of the queue. All stakeholders must agree to publishing their datasets to participate.

Generated Data At minimum, the CubeSat must generate the Demonstration Trajectory dataset. Each Trajectory Proposal submitted by a stakeholder will ideally generate a new dataset.

The datasets will be directly valuable for trajectory and pose estimation research. If enriched with post hoc annotations, perhaps derived from open-access CubeSat designs and health telemetry, they could also be used for object segmentation and self-inspection.

The minimum quality metrics of the data generated over the entire CubeSat lifespan are:

1. Absolute timestamp accuracy: This needs a robust real-time clock and periodic calibration with time servers to account for drift.
2. Pose ground truth accuracy: This is a design requirement. A Camera Payload with low pose accuracy is not suitable to generate the desired datasets and shouldn't even be launched under the this ConOps.
3. Pose diversity across datasets: Ideally this arises from the participation of diverse stakeholders. If this is not achieved, the CUBICS team can specify its own TPs to address this.

Payload The proposed payload is a robotic arm attached to the CubeSat that can be manipulated to "orbit" a camera around it as if it was a spacecraft engaging in proximity operations. This proposal is inspired by papers [9, 12] that attach cameras and CubeSat mockups to industrial robot arms to generate trajectory and pose estimation datasets in the laboratory. Image data will come from the camera. Pose ground truth data will come from the encoders on the robotic arm. Timestamps will come from a real-time clock.

Technical Challenges

- Regardless of how the camera is navigated around the CubeSat, the volume of data will be in the gigabytes. The TP queue will probably remain empty most of the time, so the CubeSat could use that idle time to downlink the dataset over several orbits.
- There are papers [7, 5, 6] describing robot arms with extensions of 30-60cm that fit within 1U, yet their controllability with inverse kinematics and capacity to maneuver a camera must be established.
- The desired degrees of freedom will require the integration of multiple motors, drawing ample energy. This could be managed by performing the smoothest transitions between poses allowed by the specified delays.
- Due to the robot arm extension and energy availability constraints, the generated datasets risk not achieving a high degree of pose variability.

Success Criteria

1. 360 degree orbiting around the CubeSat must be achievable for full success. One hemisphere would be enough for full success.
2. Images that show the CubeSat fully within the frame at every angular position mean full success. Partial CubeSat visibility controllable via camera pitch, roll, and yaw values is a partial success.
3. A minimum delay between poses that is 80% caused by camera stabilization (to avoid blur) is a full success. Anything less than that has to be contrasted against the latency of relevant navigation systems to determine a partial success.

Justification Canada will remain a leading spacefaring nation, establishes the Canadian Space Agency in its Space Strategy for Canada, by participating in the next effort to push humanity into space: the Lunar Gateway Mission [8]. Building on the success of the Canadarms, Canada's participation will materialize as a next-generation artificial intelligence-enabled deep-space robotic system for the Lunar Gateway Mission, a milestone that will allow Canada to strengthen its position as a world leader in AI-enabled space robotics and position Canadian firms as global leaders in the future on-orbit servicing market [8].

On-orbit servicing, in particular, is embedded within one ambitious future concept of operations for the Lunar Gateway where it interacts with other spacecraft and their payloads for applications including refueling, inspection, assembly, and maintenance [15, 3]. In such scenarios, servicing is generally thought to be preceded by rendezvous and proximity operations, and may be both preceeded and succeeded by a capture/release phase like docking, grappling or berthing [2]. Proximity operations is a particularly challenging phase that in turn comprises the awareness and navigation capabilties leveraged when two spacecraft operate close to each other [4, 12, 9, 16].

The research community argues that awareness in proximity operations, the capability supported by this project, should be adaptable to targets ranging from cooperative spacecraft to non-cooperative debris. This must be accomplished without compromising robustness to the tight energy constraints of spaceborne systems and the ample variations in space conditions such as lighting and electromagnetic noise [16, 12, 9, 19].

Cameras, particularly the common monocular, red-green-blue light intesity sensing type, have been put forward as the sensor of choice for proximity operations awareness owing to their relative energy efficiency, high frame rate, and low cost [12, 9]. Space institutions have consequently challenged researchers to exploit images as the input for awareness problems like pose estimation and object segmentation, as evidenced by the 2019 and 2021 editions of the Satellite Pose Estimation Competition hosted by NASA and ESA [12], and NASA's 2024 Pose Bowl Spacecraft Detection and Pose Estimation Challenge [16]. The academic community has applied solutions including the feature-based Optical Flow, Structure from Motion [4], and Scale Invariant Feature Transform (SIFT) [4], and varied [12, 9, 16] Convolutional Neural Networks (CNNs) trained on large volumes of annotated images, all of which are robust to non-cooperative targets.

Researchers have concluded that the quality of images available to tune or train the pose estimation and object detections models is a significant predictor of their performance, where quality in this domain is determined by whether the images are synthetic, lab-generated, or orbital [12, 9, 16]. Identified shortcomings of synthetic datasets include not resembling the visual features [12, 9] and variability in illumination [12, 9] of orbital images. Interestingly, one paper added more than thirty thousand synthetic images to a spacecraft segmentation dataset and obtained a negligible improvement in performance [16].

Authors have thus turned to enriching image datasets with images generated in the laboratory. Setups with spacecraft mockups and cameras mounted on robotic arms plus light boxes and sun lamps as illumination sources have been used to generate pose ground truth labels for individual poses [12] and predefined, controlled time series with many degrees of freedom that simulate the trajectories of spacecraft engaged in proximity operations [9]. These experimental pose annotations and time series of images have been leveraged to measure the domain gap of pose estimation models trained entirely on synthetic data, which can be similar when using either lab-generated or orbital images as the reference [12], and to estimate trajectories as the change in pose over time using Temporal Convolutional Networks [9], respectively.

However, lab-generated images have been deemed insufficient for validation of pose estimation under real space conditions [14], even by the publishers [11]. Criticisms and limitation acknowledgements include unrealistic illumination conditions [19, 12] and materials [12, 9], and the presence of

visual artifacts [12]. Lab-generated dataset quantity is still a concern, as authors have called for the publication of more labelled sequences for realistic proximity operations trajectories [1], ensuring that all data points include pose labels, not just a few hundreds [12, 9, 17], and maintaining a high pose diversity [9].

Orbital image datasets suitable for pose estimation are extremely rare, as the only publicly-accesible example found was PRISMA25, of only 25 images, generated during a proximity operations demonstration mission [13]. A dataset of 540 orbital images of Cygnus spacecraft with manual pose annotations was identified, but it is privately held [9]. Spacecraft segmentation can leverage orbital imagery from space agency and internet archives estimated to be in the few thousands, yet substantial human annotation effort is often necessary [12, 16], and segmentation mask quality is varied [16]. No orbital datasets fo trajectory estimation were identified.

Other orbital datasets with interesting technological enablers include two examples with cameras mounted on a tethered daughter satellite [10] and an extendable boom on a CubeSat [18], respectively. They originally supported self-inspection, as in tether deployment validation [10] and material degradation and micrometeoroid impact sensing [18]. Images generated with the boom-mounted camera were annotated only with the corresponding extension of the boom [18], while the tethered camera images had no labels [10]. No further application of either dataset was found.

Relevantly, the ISS Selfie initiative mounted a camera on the International Space Station's (ISS) Canada-made Canadarm2 and Dextre robot arms [19] to generate more than 49 hours of proximity operations simulation video with time and pose telemetry. This data was used to demonstrate the validation of SIFT-based pose estimation algorithms, and thus serve as a precedent for future on-orbit services that adopt the International Docking System Standard interface [19].

An analysis of the citations of the ISS Selfie publication found no subsequent applications of its dataset. It can be speculated that this is in part due to it not being transferable to the CubeSat form factor on which space research is primarily carried out.

In synthesis, trajectory estimation, pose estimation, object segmentation, and self-inspection are problems in awareness for proximity operations related by their vision-based approaches. Self-inspection needs orbital images, which can be analyzed post hoc. Object segmentation needs orbital images labelled with segmentation masks, which can be annotated post hoc. Pose estimation needs orbital images labelled with pose ground truth data, which are extremely rare. Trajectory estimation needs orbital image time series labelled with pose ground truth data, of which no examples were found. All need diverse orbital datasets with sizes at least in the tens of thousands.

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